FLUTTER DEV · ON-DEVICE AI

ARSENIUS PURBANDONO

I write Flutter apps for tax officers in West Java. Lately I've been teaching those apps to read license plates on the phone itself, no internet needed.

Bandung, Indonesia · West Java Revenue Agency (Bapenda Jabar) · since Feb 2023

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THE NUMBERS

3+ years writing Flutter at work	**56→76%** plates read exactly right, ML Kit vs my model	**−30MB** smaller APK after I removed ML Kit
4× faster crop after moving it off the UI thread	**1.1M** parameters in the OCR model (2.3 MB file)	**494** test plates I locked away before training

HOW THE PHONE READS A PLATE

This runs inside Panah Pasopati. Everything happens on the

phone. No server, no signal needed.

01
CAMERA
Frames from a cheap or mid-range Android phone

02
DETECT
YOLOv11n finds the plate and straightens it

03
CROP
Cropped once, in a background isolate

04
COLOR
MobileNetV2 tells white, black, yellow and red plates apart

05
READ
My CNN+CTC model reads the characters

06
LOOKUP
Number + color go to the vehicle tax lookup

APPS

Panah Pasopati
Live · Flutter · TFLite · YOLO · Shorebird · Firebase · Kotlin

Officers point the camera at a plate and get the vehicle's tax record. Used in the field across West Java.

- Built the plate-recognition module (Teman Pamor): detection, color, OCR, all on the phone
- Patched the Android camera plugin in Kotlin so the camera stops restarting 3–4 times
- Locked down OTA updates: HTTPS only, newer versions only, SHA-256 check

SAKTI JAWARA
Live · Flutter · Device SDK · Thermal printer · Excel/PDF

The counter app at every Samsat office in West Java. Officers take tax payments and change vehicle data here.

- Hooked up the device SDKs: camera, thermal printer, second screen
- Excel and PDF export
- Keyboard shortcuts, because counter staff hate reaching for the mouse

ATOS PAMOR
Live · Flutter · MVC · Dio · Hive · flutter_map · Geolocator · OTA

The heavier tracing app for the special task force: data checks, vehicle history and the paperwork for collecting overdue tax.

- Photo evidence gets a watermark with officer ID, GPS and the street address
- Every controller handles errors the same way now, and users see the real backend message
- Moved the app to a new major Flutter and Dart version

Sakti Mobile
Live · Flutter · Camera · Excel/PDF

The phone version for verifiers who check payments and data changes.

- Camera for evidence photos
- Local data storage
- Excel and PDF export

Samsat Digital
Live · Flutter · Kiosk · Device SDK

Self-service kiosk at Samsat branches where people pay vehicle tax on their own.

- Connected the kiosk hardware: camera, printer and the rest
- Local data for each kiosk session

SEMAR

`Live · Flutter · Android + Web · Multi-package · Hive`

Water tax. Officers record usage in the field, a verifier approves it.

- Split into packages: recording, verification, history
- One codebase for Android and the web
- Works offline first, syncs later (Hive)

Pajak Alat Berat

`Live · Flutter · MVC`

Tax app for heavy equipment like excavators and cranes: recording and tracing.

- Features and maintenance on the shared Flutter setup

Aing Linmas

`Prototype · Flutter · Android + iOS · QR · PDF`

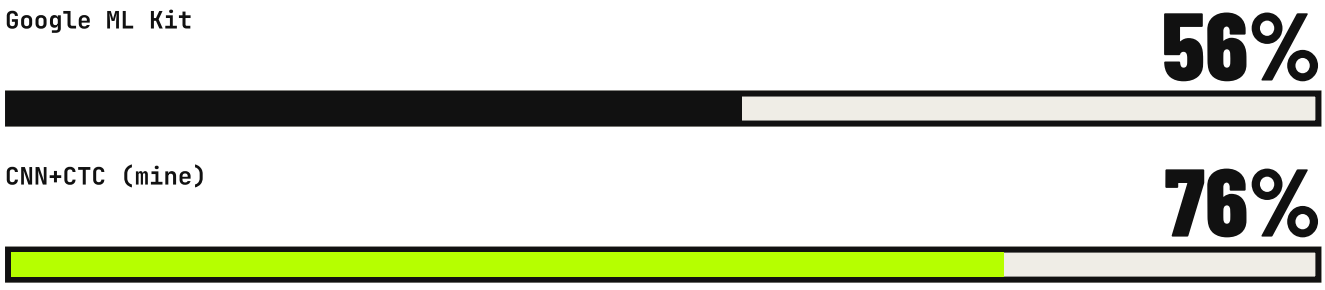
Phone version of a sign-up system for Linmas community-protection members, from province down to village.

- 3-step sign-up with an ID-number check and drafts that save themselves
- Digital member card with a QR code, saves as image or PDF
- Runs on fake data for now, no backend yet

MODEL LAB

`The plate-recognition repo: data, labels, training, and the test set I'm not allowed to peek at.`

WHY I DROPPED GOOGLE ML KIT



ML Kit is a general OCR. On Indonesian plates it kept mixing up O and 0, so officers had to fix the result by hand.

First I set aside 494 real plates as a test set and split by plate, so the same car never shows up in training. Then I generated 30k synthetic plates in the plate font, trained a small CNN+CTC model and exported it to TFLite, checking that the outputs still matched.

It only went into the app after it beat ML Kit on that test set: 76% vs 56% exact match. Dropping ML Kit also took about 30 MB off the APK.

Plate detector

LIVE

YOLOv11n → TFLite

mAP50 81.6%

Finds the plate and fixes the rotation when the photo is taken at an angle.

Plate OCR

LIVE

CNN + CTC · 1.1M params · 2.3 MB

76% exact

Reads the characters. ML Kit got 56% on the same 494 plates.

Plate color

LIVE

MobileNetV2 α0.35 · 96×96

~97% val acc

White, black, yellow, red. Red is still weak: not many real red plates to train on.

SANDBOX

Side projects I did to learn NLP and transfer learning.

Amazon Review Sentiment

87.25% acc · F1 0.87

TF-IDF · Logistic Regression · NLTK · Flask

1.6M reviews to train, 400k to test.

Casting Defect Inspection

91.8% val acc

MobileNetV2 transfer learning · Keras · Tkinter

Good vs defective pump impellers, 1,016 factory photos.

WORK LOG

WORK

a76c0de Mobile Application Developer

HEAD

West Java Revenue Agency (Bapenda Jabar) · Feb 2023 – now

Flutter apps for vehicle and tax tracing, Samsat counters and field checks. Codebases split into 4–19 packages, MVC and Clean Architecture, updates through Shorebird and OTA. I also trained and shipped the plate-recognition models.

5a1e922 Mobile Developer Intern

PT Safepedia Global Teknologi, Jakarta · Aug – Dec 2022

Native Android in Kotlin for an app used by PT Antam's mining operators. Bug fixes and new features.

SCHOOL

Information Systems & Technology Education, Universitas Pendidikan Indonesia (2019–2023)

CERTIFICATES

- **BNSP Senior Mobile Programmer** · 2025-11
- **BNSP Junior Mobile Programmer** · 2025-08
- **EF SET English B2** (54/100) · 2024-06
- MSIB Mobile & Cloud Dev, Nongsa Digital Park · 2022
- MSIB Game Development, Agate · 2021–22

LANGUAGES

- Japanese: beginner. Working through Minna no Nihongo, aiming for JLPT N4
- English: B2 (EF SET 54/100)
- Indonesian: native

I want to work as a software or AI engineer in Japan. I can start in the second half of 2027 and will need visa sponsorship.

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